SafeView

Intelligent Image Content Moderation Tool

COMP 264 — Cloud Machine Learning | Deliverable 5: Component Interaction Diagrams

Diagram 1 — Primary Flow: POST /analyze

Traces all 17 messages across the five named components during a full image moderation request. Solid arrows are requests/calls; dashed arrows are responses/returns; dotted arrows are conditional calls (only fired when text is detected in the image). Activation bars show which components are active at each step.

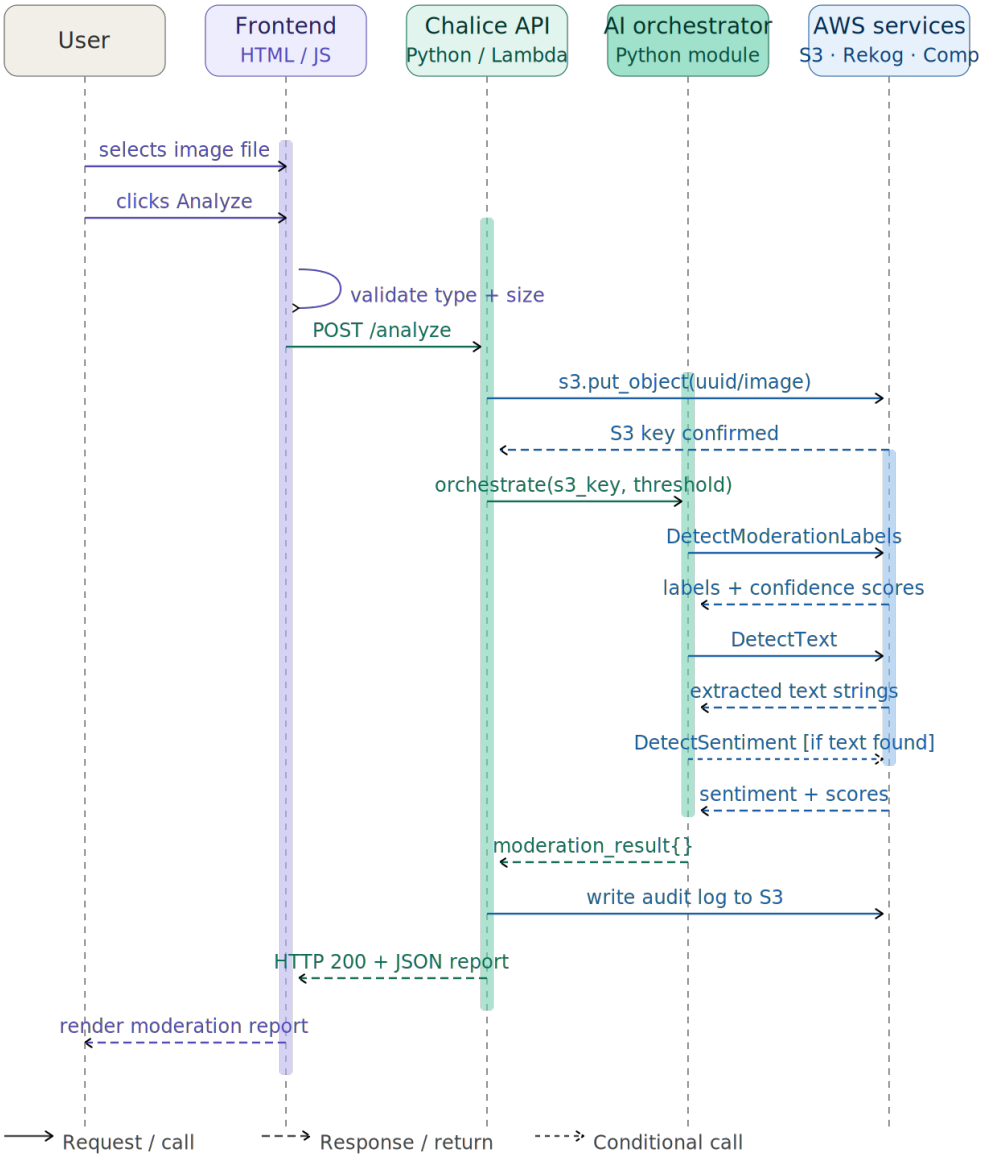

Figure 1: Full interaction sequence for UC-01 (Upload Image for Moderation) — User → Frontend → Chalice API → AI Orchestrator → AWS Services

Diagram 2 — Threshold Re-Analysis Flow: POST /analyze/threshold

Shows the short-circuit path triggered when the user adjusts the confidence threshold slider (UC-03). The AWS services component is greyed out — no Rekognition, Comprehend, or S3 calls are made. The AI Orchestrator

re-filters the cached result in memory and returns an updated verdict.

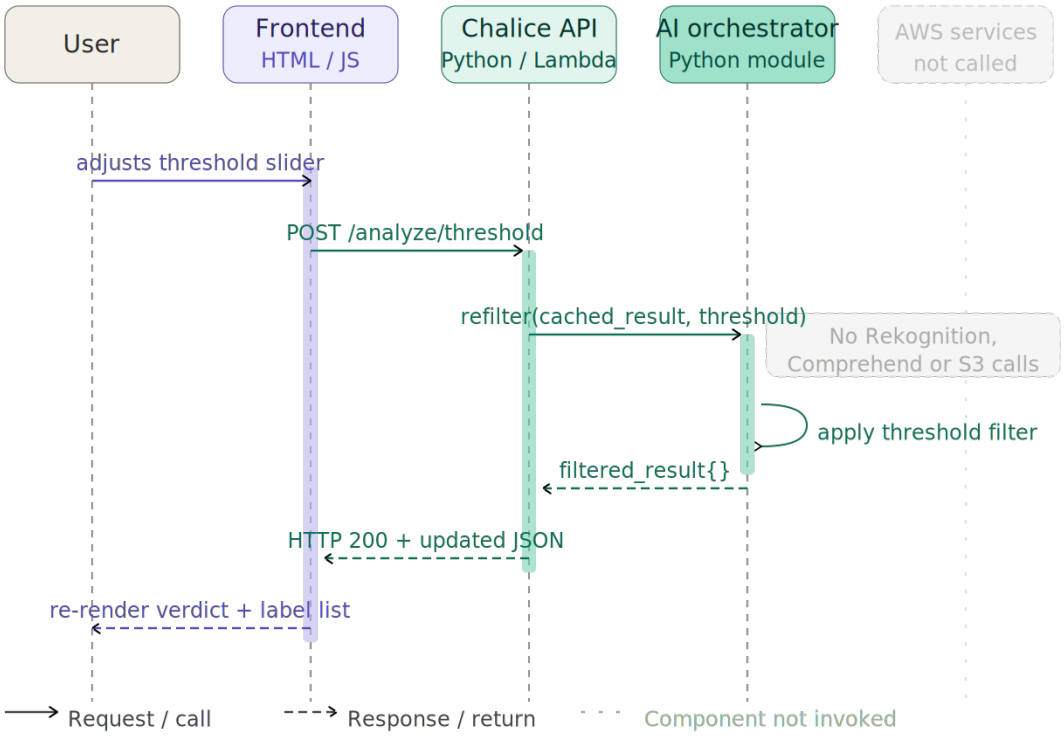

Figure 2: Threshold re-analysis sequence for UC-03 — re-filters cached results without any new AI service calls

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Component key

- **User:** Platform operator or developer using the SafeView web interface
- **Frontend (HTML/JS):** Single-file web interface — handles file selection, validation, rendering, and threshold slider
- **Chalice API (Python/Lambda):** REST endpoint handler — routes requests, calls S3, delegates to AI orchestrator, writes audit log
- **AI orchestrator (Python module):** Isolated module that calls Rekognition and Comprehend via Boto3 and applies threshold filtering
- **AWS services (S3 · Rekognition · Comprehend):** Managed AWS AI and storage services — called only during POST /analyze, never during threshold re-analysis